

```
In [1]: #RISHAB GANESH N   CODETECH_DATASCIENCE internship ( TASK -1) 03/02/26
#Import Required Libraries
```

```
import pandas as pd
import numpy as np

from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
```

```
In [2]: #Create / Load Dataset
```

```
raw_df = pd.DataFrame({
    "Age": [22, 25, 30, None, 28],
    "Salary": [30000, 45000, 50000, 42000, None],
    "Department": ["IT", "HR", "IT", "Finance", "HR"],
    "Experience": [1, 3, 5, 4, None]
})

raw_df.to_csv("../data/raw_data.csv", index=False)
```

```
In [3]: #Load Dataset
```

```
df = pd.read_csv("../data/raw_data.csv")
df.head()
```

```
Out[3]:
```

	Age	Salary	Department	Experience
0	22.0	30000.0	IT	1.0
1	25.0	45000.0	HR	3.0
2	30.0	50000.0	IT	5.0
3	NaN	42000.0	Finance	4.0
4	28.0	NaN	HR	NaN

```
In [4]: #Data Inspection
```

```
df.info()
df.isnull().sum()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Age         4 non-null     float64
1   Salary      4 non-null     float64
2   Department  5 non-null     object
3   Experience  4 non-null     float64
dtypes: float64(3), object(1)
memory usage: 288.0+ bytes
```

```
Out[4]:
```

	Age	Salary	Experience
count	4.00	4.0	4.000000
mean	26.25	41750.0	3.250000
std	3.50	8500.0	1.707825
min	22.00	30000.0	1.000000
25%	24.25	39000.0	2.500000
50%	26.50	43500.0	3.500000
75%	28.50	46250.0	4.250000
max	30.00	50000.0	5.000000

```
In [5]: #Separate Column Types
```

```
num_cols = df.select_dtypes(include=["int64", "float64"]).columns
cat_cols = df.select_dtypes(include=["object"]).columns
```

```
In [6]: #Build Numerical Pipeline
```

```
numeric_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="mean")),
```

```
    ("scaler", StandardScaler())
])
```

In [7]: *#Build Categorical Pipeline*

```
categorical_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("encoder", OneHotEncoder(handle_unknown="ignore"))
])
```

In [8]: *#Combine Pipelines*

```
preprocessor = ColumnTransformer([
    ("num", numeric_pipeline, num_cols),
    ("cat", categorical_pipeline, cat_cols)
])
```

In [9]: *#Apply ETL Pipeline*

```
processed_data = preprocessor.fit_transform(df)
processed_data
```

```
Out[9]: array([[ -1.56763612, -1.78460997, -1.70084013,  0.        ,  0.        ,
                  1.        ],
               [ -0.46106945,  0.49361552, -0.18898224,  0.        ,  1.        ,
                  0.        ],
               [  1.38320834,  1.25302402,  1.32287566,  0.        ,  0.        ,
                  1.        ],
               [  0.        ,  0.03797042,  0.56694671,  1.        ,  0.        ,
                  0.        ],
               [  0.64549722,  0.        ,  0.        ,  0.        ,  1.        ,
                  0.        ]])
```

In [10]: `np.save("../data/processed_data.npy", processed_data)`

In [11]: `import os`
`os.listdir("../data")`

Out[11]: ['processed_data.npy', 'raw_data.csv']

In []: